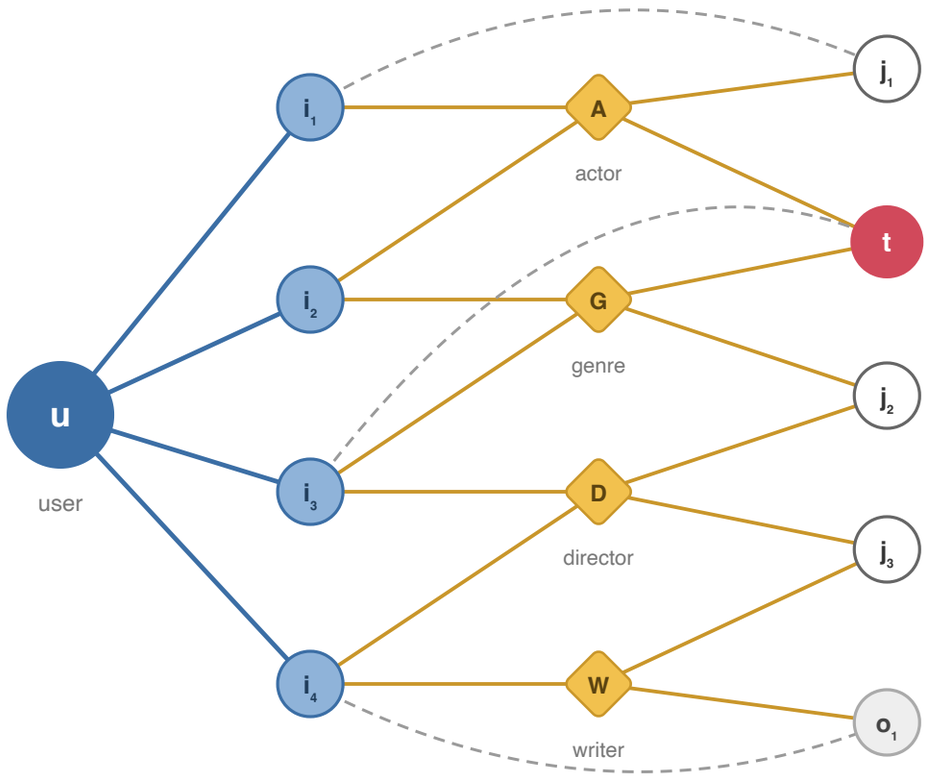
a Original recommendation



score & rank

Top-K list

rank	
1	j_1
2	t
3	j_2
4	j_3
⋮	
10	j_9
K = 10	
	o_1 outsider

KG path · KG entity · collaborative pa · target t

Request: remove t, keep the rest

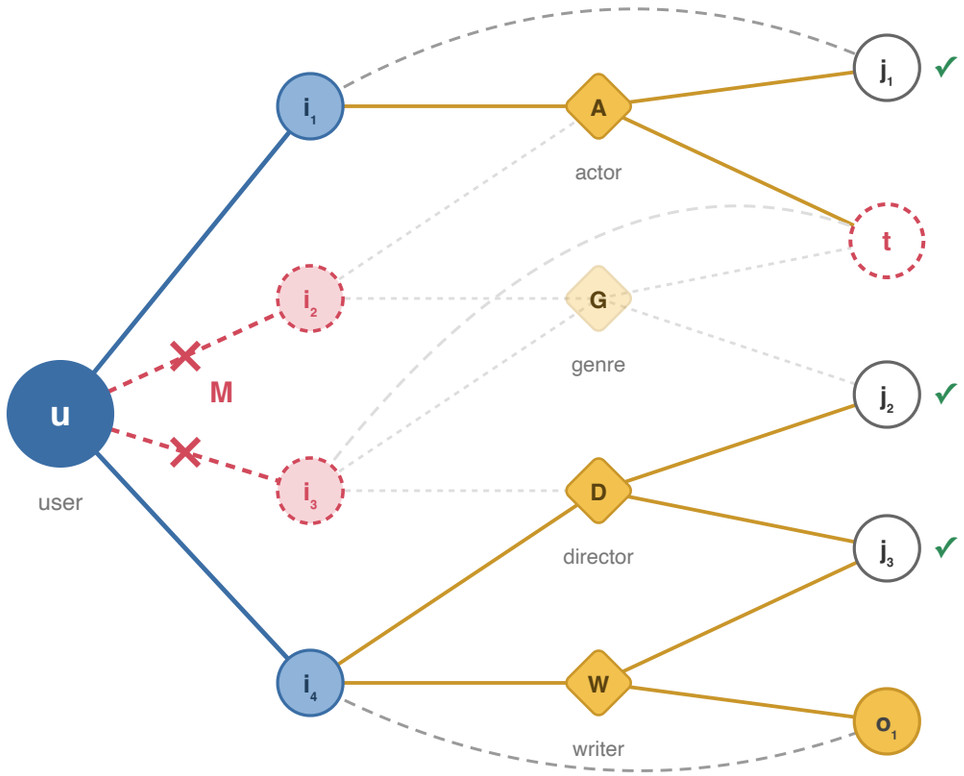
- **C1** t leaves top-K
- **C2** other K−1 items stay
- **C3** each keeps ≥1 evidence path

min |M| s.t. C1+C2+C3

exact MILP · lazy constraints
every mask verified by re-ranking

apply M

b Counterfactual with mask M = {i_2, i_3}



re-rank

Counterfactual top-K

rank		
1	j_1	✓
2	j_2	✓
3	j_3	✓
⋮		
9	j_9	✓
10	o_1	new
K = 10		
	t	removed

Outcome

- **C1** t removed ✓
- **C2** j_1 … j_9 remain; o_1 enters ✓
- **C3** j_1 via A · j_2 via D · j_3 via W ✓

Explanation: "without i_2 and i_3, t would not be recommended"